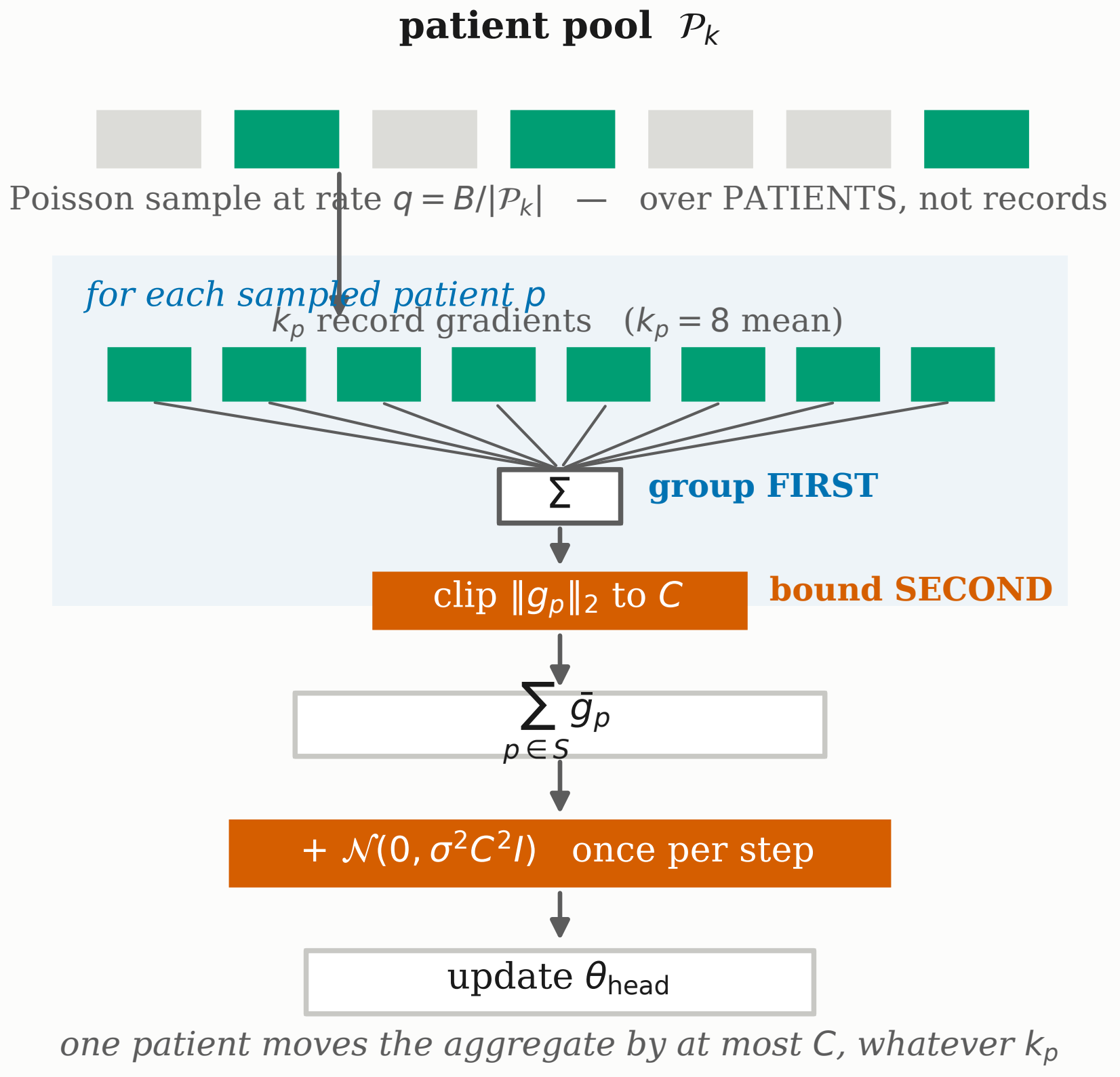
patient pool $\mathcal{P}_k$
Poisson sample at rate $q = B/|\mathcal{P}_k|$ — over PATIENTS, not records
for each sampled patient p
k_p record gradients ($k_p = 8$ mean)
Σ
group FIRST
clip $\|g_p\|_2$ to C
bound SECOND
$\sum_{p \in S} \bar{g}_p$
$+ \mathcal{N}(0, \sigma^2 C^2 I)$ once per step
update θ_{head}
one patient moves the aggregate by at most C, whatever k_p